

Jacob Lynn

 Jacob Lynn |  jacob.lynn280@gmail.com |  +949-691-4585
<https://jacobnlynn.github.io>

EDUCATION

Stanford University

Expected Graduation 2029

B.S. Mechanical Engineering

Minor in Computer Science

GPA: 3.41

Relevant Coursework:

CS223A: Intro to Robotics, ME210: Intro to Mechatronics

RELEVANT EXPERIENCE

ARMLab Researcher - Assistive Robotics and Manipulation

Apr 2026 - Present

- Designed an electromagnetic linear actuator for a robotic muscle for tendon-driven robot hands with full position (PID) and force control; patent application filed (patent pending)
- Coded ROS2 environments for robot finger and hand control and a force simulation to determine electromagnet spacing using data I collected
- Designed a sensor that calculates the position of the actuator to less than a millimeter by measuring the changing inductance as permanent magnets move through it
- Prototype weighs 70g and delivers $\sim 5\text{N}$ (500g-force) continuously over 15mm stroke, a $\sim 7:1$ force-to-weight ratio
- Designed for a 16 actuator array (each powering a finger joint) to control a full dexterous robot hand
- Worked 40hr weeks for 10 weeks over the summer and continuing in the lab during the school year

Stanford Student Space Institute - Europa Team Colead

Sep 2025 - Present

- Co-Managed a team of ~ 10 people to design a low-cost, $< \$5\text{K}$, open source benthic lander (designed for ocean floor depths) that collects sediment, water, DNA, and video
- Led the mechanical design, which includes a custom soil scoop, Niskin tube, and a pressure-powered DNA collector
- Ran 4 successful field tests in SF Bay and Monterey Bay ($\sim 100\text{m}$) with MBARI; collected sediment samples, video, and GPS tracking data

Stanford Student Space Institute - IREC Team, Payload Subteam

Sep 2025 - Present

- Led the redesign of an L3 rocket payload that synthesizes crystals in microgravity for pharmaceutical testing to compete in the International Rocket Engineering Competition (IREC)
- Engineered a rigid, compact payload and vibration-proof interior structure that fit within a 2U CubeSat volume that flew and formed part of the structure on a rocket that went $> \text{Mach } 1$

PROJECTS

Modular Robot Dog

[Link to project](#)

- Designed and tested a custom QDD actuator with a 5:1 ratio to power a robot dog with $\sim 4\text{Nm}$ of continuous torque at $\sim 740\text{RPM}$
- Built a full-sized 12DoF robot dog standing $\sim 6\text{in}$ off the ground; currently working on the electronics of it
- Created a modular system of actuators to act as a framework for limbed robots
- Used system to design 7DoF Robot arms, multiple sizes of robot dog, and a mimic of JPL's RoboSimian

SKILLS

Engineering

CAD, CAM (Fusion 360, Onshape)

Fabrication

CNC machining, 3D printing, laser cutting, carbon fiber/composite epoxy layups

Programming

Python, JavaScript, Java, C, C++, ROS2, Linux

Control/Embedded Systems

CAN, microcontrollers (Teensy 4.1, Arduino Uno, ESP32), PID, KiCad